

IMPACT pSIDDHI 3.0

Final-Term Submission Document

Common template for all tracks (Custom · Data · Platform) and all semesters, including pSiddhi Lite | Covers the ENTIRE programme, Week 4 through Week 17

1. Participant & Project Identification

Topic ID (as finalised by L&D)	S4-I-18
Topic Title	Smart Facility Management Platform
Participant Name	Subhiksha Thirunavukkarasu
Employee ID	P399
Track	☒ Custom ☐ Data ☐ Platform
Semester & Category	Semester 4 — Integration Mastery (Capstone): E2E Integration with IoT, ML/AI Core, Azure Deployment, QA Mandatory
Participation Type	☒ Regular ☐ pSiddhi Lite
Approved Budget Ceiling	₹2,500 (fixed)
Mid-Term document filename (as uploaded to Moodle)	S4-I-18_Subhiksha_Thirunavukkarasu_MidTermDoc.docx
Mid-Term result / feedback received	☐ On track ☐ At risk — actions were assigned ☒ Other: official result/feedback not supplied
Final Review Window	Week 17 — submission due 21-Sep-2026; exact review dates not supplied

2. Approved Proposal Recap

2.1 Problem Statement (as approved)

No change from Mid-Term recap. Facility teams operate reactively because available HVAC, electrical and occupancy telemetry is not converted into decisions. FacilityIQ addresses preventable failures, invisible space use, energy waste and calendar-based maintenance through integrated telemetry, ML scoring and an operational portal.

2.2 Proposed Solution Summary (as approved)

FacilityIQ implements simulated three-domain devices, unified validation, a JSONL data lake mirrored to Azure Blob, feature pipelines, three scikit-learn models and a five-screen Streamlit portal. The direct-to-Blob path remains the working Azure fallback because corporate RBAC blocked IoT Hub, Functions and App Service provisioning. Final QA adds Great Expectations, Selenium browser journeys and Locust load testing. Three report-ready datasets were validated and used to create live Power BI Service reports for equipment health, energy consumption and space utilization.

2.3 Core Tools & AI Components (as approved)

Azure IoT Hub, Azure Functions, Azure Blob Storage, Azure App Service; Databricks Community and MLflow; scikit-learn and PyCaret; Streamlit; Power BI Desktop; Gemini; Ollama/Llama; GitHub Actions; Pytest, Selenium and Great Expectations. Actual usage and substitutions are reconciled in Section 7. The supplied project record also includes Groq, but the official L&D feedback that allegedly required it was not supplied for independent

verification.

3. Progress Against Approved Plan (Full Programme: Week 4-17)

ID	Planned Deliverable (per approved proposal)	Planned Window	Carried from Mid-Term? (Y/N)	Status	Evidence ID(s)
D-01	Azure setup, telemetry schema and HVAC simulators	Week 4	Y	Partial	EV-04, EV-06 (Mid-Term)
D-02	Energy/space simulators and 3-domain landing zone	Week 5	Y	Done	EV-04, EV-06 (Mid-Term), EV-07
D-03	Lake-to-analytics feature pipeline	Week 6	Y	Done	EV-04, EV-05 (Mid-Term), EV-07
D-04	HVAC Random Forest trained and measured	Week 7	Y	Done	EV-04 (Mid-Term), EV-09
D-05	Electrical Isolation Forest and motor GBM	Week 8	Y	Done	EV-04 (Mid-Term), EV-09
D-06	Portal health overview and equipment detail	Week 9	Y	Done	EV-01-EV-03 (Mid-Term)
D-07	Embedded unit/integration/model/UI QA and CI	Weeks 4-9	Y	Done	EV-05 (Mid-Term), EV-08
D-08	Role-based login, maintenance and AI screens	Weeks 11-12	Y	Done	EV-01, EV-03 (Mid-Term)
D-09	Power BI dashboards: equipment, energy and space	Week 13	N	Done	EV-11
D-10	Scale and browser E2E validation	Week 14	N	Done	EV-08
D-11	Great Expectations data quality and full regression	Week 15	N	Done	EV-07, EV-08
D-12	Final documentation, evidence and review preparation	Week 16	N	Done	EV-07-EV-11

3.1 Overall Final Self-Assessment

RFP-defined final checkpoint / definition of done (summarise in 2-3 lines)	Complete three-domain ingestion, three prediction scenarios, four portal screens, three analytics dashboards, Azure deployment, at least 10,000 telemetry points and a QA suite with at least 80% measured coverage.
% of overall project completed (honest estimate)	88% — core telemetry, models, portal, AI, QA and three Power BI Service reports are working and evidenced. The corporate-access-blocked IoT Hub/Functions/App Service route prevents a 100% claim.
% reported at Mid-Term (from your Mid-Term Section 3.1)	90% of the Week-10 checkpoint (not 90% of the full programme).
Is the final solution demonstrable live, end-to-end, at the review?	☐ Yes, end-to-end ☒ Yes, partially ☐ No, recording/screenshots only

4. Evidence Pack (Full Programme: Week 4-17)

4.1 Evidence Index

Evidence ID	Caption — what does this prove?	Deliverable ID(s)	Verifiable link (if any)	Carried from Mid-Term? (EV-ID if yes)
EV-01	Role-based login and admin panel	D-06, D-08	https://github.com/subhikshathirunavukk arasu0503/facilityiq	Yes — Mid-Term EV-01
EV-02	Health overview, ranked alerts and live context	D-05, D-06	https://github.com/subhikshathirunavukk arasu0503/facilityiq	Yes — Mid-Term EV-02
EV-03	Equipment, maintenance and AI portal screens	D-06, D-08	https://github.com/subhikshathirunavukk arasu0503/facilityiq	Yes — Mid-Term EV-03
EV-04	Three held-out ML model result sets and lake scale	D-01-D-05	https://github.com/subhikshathirunavukk arasu0503/facilityiq/tree/main/models	Yes — Mid-Term EV-04
EV-05	Measured Mid-Term coverage and green CI	D-03, D-07	https://github.com/subhikshathirunavukk arasu0503/facilityiq/actions	Yes — Mid-Term EV-05
EV-06	Azure Blob lake and repository history	D-01, D-02	https://github.com/subhikshathirunavukk arasu0503/facilityiq/commits/main	Yes — Mid-Term EV-06
EV-07	Great Expectations: 14,784 records, 51/51 checks, 0 duplicates	D-02, D-03, D-11	https://github.com/subhikshathirunavukk arasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final	No — new
EV-08	80-test regression, 87% coverage, Selenium 2/2, Locust 1,402 requests/0 failures	D-07, D-10, D-11	https://github.com/subhikshathirunavukk arasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final	No — new
EV-09	Final model metrics and honest completion boundary	D-04, D-05, D-12	https://github.com/subhikshathirunavukk arasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final	No — new
EV-10	Final source/evidence commit and Power BI datasets	D-09-D-12	https://github.com/subhikshathirunavukk arasu0503/facilityiq/commit/ae4a8b4df6493998431db7f1c29786b9c15fd6fa	No — new
EV-11	Three live Power BI Service reports and semantic models	D-09	https://github.com/subhikshathirunavukk	No — new

Evidence ID	Caption — what does this prove?	Deliverable ID(s)	Verifiable link (if any)	Carried from Mid-Term? (EV-ID if yes)
			arasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final	

4.2 Evidence Blocks (paste screenshots here)

EV-07 — Final telemetry data-quality gate

What this proves	Great Expectations validated every committed lake record: 195 files, 14,784 records, 0 duplicate device/timestamp pairs and 51/51 domain expectations passed.
Deliverable ID (Sec 3)	D-02, D-03, D-11
Date of Development / Testing	20-Sep-2026
Verifiable link	https://github.com/subhikshathirunavukkarasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final
Carried from Mid-Term?	☒ No (new / progressed)

```

Great Expectations - final telemetry quality gate

Run: 20-Sep-2026 | source: committed data/lake
195 JSONL files | 14,784 telemetry records | 0 duplicate device/timestamp records
HVAC      6,720 records | 19/19 expectations passed
Energy    5,376 records | 15/15 expectations passed
Occupancy 2,688 records | 17/17 expectations passed
TOTAL     51/51 expectations passed | SUCCESS: true
Checks: required fields, timestamps, domains, non-null metrics, physical value ranges

```

EV-08 — Final regression, browser E2E and load run

What this proves	80 pytest tests passed with 87% statement coverage; 2/2 Selenium journeys passed; Locust produced 1,402 requests with 25 concurrent users, 0 failures and p99 5 ms on the local Streamlit HTTP surface.
Deliverable ID (Sec 3)	D-07, D-10, D-11
Date of Development / Testing	20-Sep-2026
Verifiable link	https://github.com/subhikshathirunavukkarasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final
Carried from Mid-Term?	☒ No (new / progressed)

```

● ● ● Final QA regression, Selenium E2E and Locust load run

Run: 20-Sep-2026 | Python 3.10 | local Streamlit portal
pytest full suite: 80 passed, 3 skipped | 87% measured statement coverage
Selenium browser E2E: 2/2 passed
- admin: login and navigate all five portal screens
- viewer: role restriction verified
Locust: 25 concurrent users for 30 seconds
1,402 HTTP requests | 0 failures | 47.13 requests/second
median 2 ms | p95 3 ms | p99 5 ms | maximum 10 ms

```

EV-09 — Model results and completion boundary

What this proves	Confirms current held-out model metrics and explicitly separates delivered scope from cloud components blocked during the final Azure validation attempt.
Deliverable ID (Sec 3)	D-04, D-05, D-12
Date of Development / Testing	20-Sep-2026
Verifiable link	https://github.com/subhikshathirunavukkarasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final
Carried from Mid-Term?	☒ No (new / progressed)

```

● ● ● Final model metrics and honest completion scope

HVAC Random Forest: F1 0.9965 | ROC-AUC 0.9999 | held-out n=972
Motor Gradient Boosting: F1 0.9788 | ROC-AUC 0.9990 | held-out n=972
Electrical Isolation Forest: recall 1.0000 | F1 0.7672 | held-out n=1,944
Delivered: 3-domain lake, 3 models, 5-screen portal, AI narratives, final QA
Azure Blob fallback remains; IoT Hub/Functions/App Service blocked by corporate RBAC
Power BI x3 not delivered; listed as incomplete/future scope in final submission
No blocked cloud item is claimed as completed

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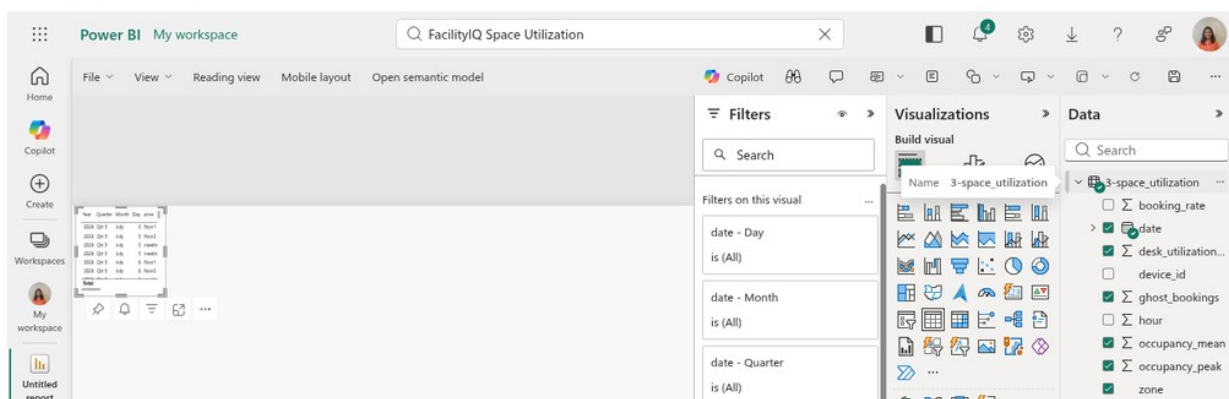
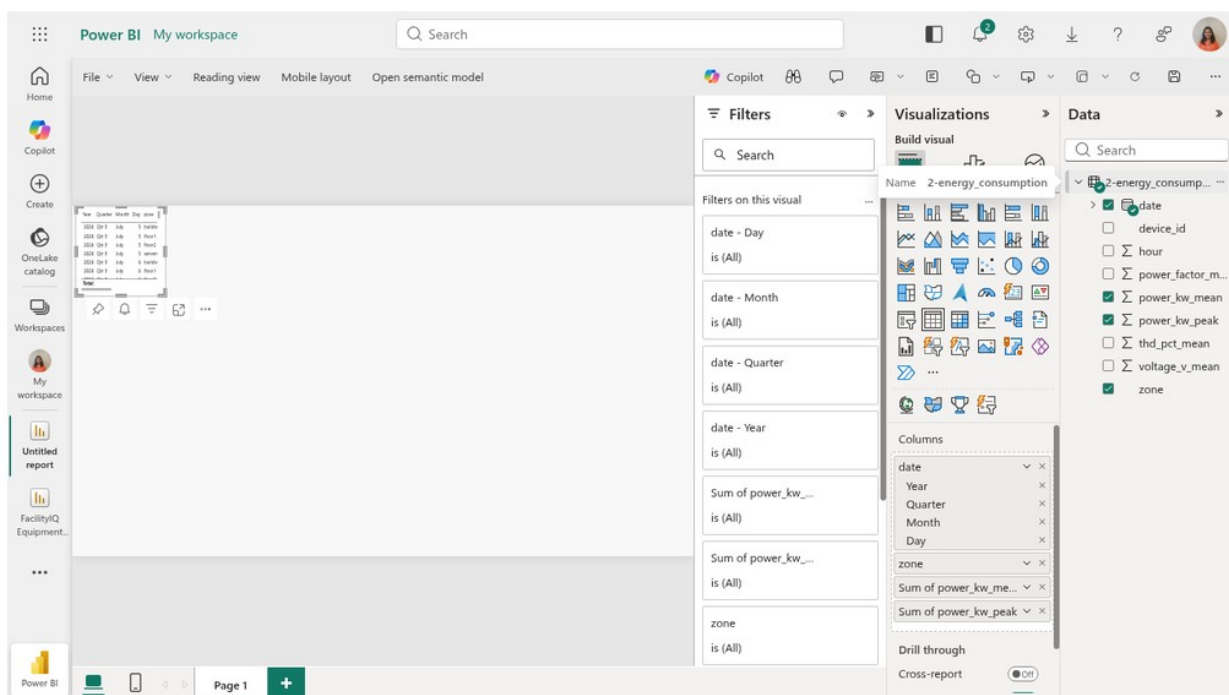
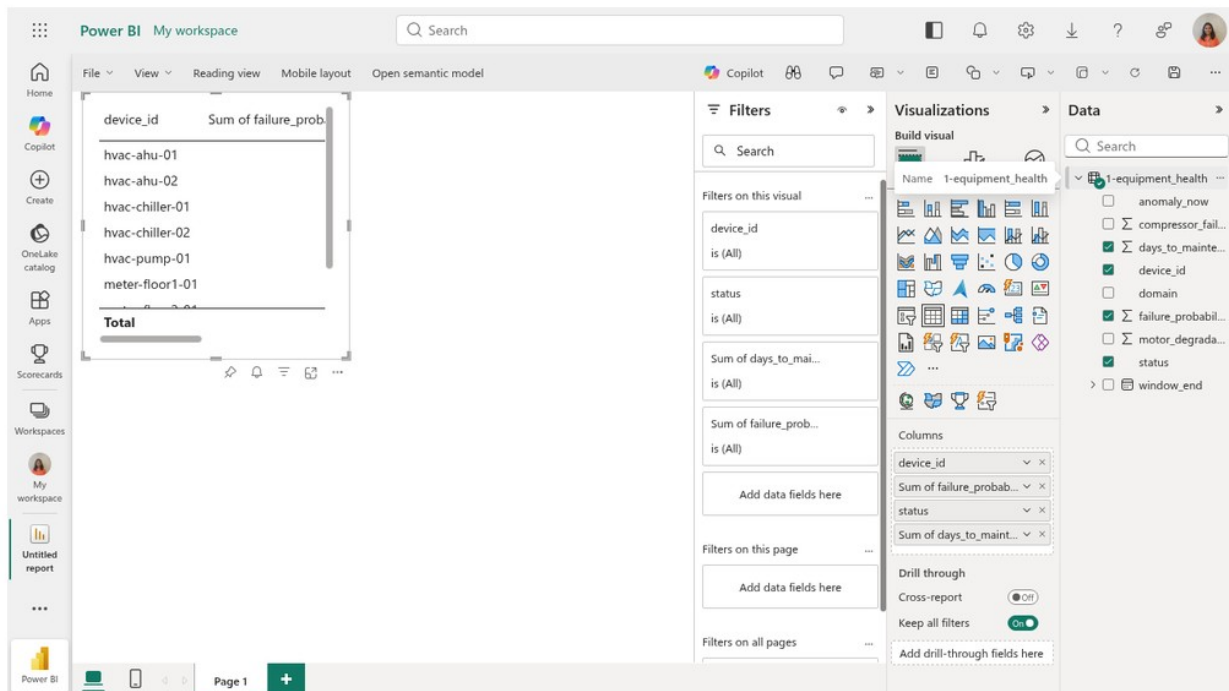
EV-10 — Final source, evidence and dataset commits

What this proves	Commits 3bac9f2 through ae4a8b4 add the Great Expectations validator, Locust profile, Selenium journeys, tests, raw result files and three validated Power BI datasets.
Deliverable ID (Sec 3)	D-10-D-12
Date of Development / Testing	20-Sep-2026
Verifiable link	https://github.com/subhikshathirunavukkarasu0503/facilityiq/commit/ae4a8b4df6493998431db7f1c29786b9c15fd6fa
Carried from Mid-Term?	☒ No (new / progressed)

EV-11 — Three Power BI Service reports

What this proves	Live reports and semantic models were created in My workspace for Equipment Health, Energy Consumption and Space Utilization. Each report contains a real table visual built from the validated CSV export.
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Deliverable ID (Sec 3)	D-09
Date of Development / Testing	21-Sep-2026
Verifiable link	Equipment: https://app.powerbi.com/groups/me/reports/ced34aee-008a-46a2-b0fa-ffb1a9b85d80?experience=power-bi ; Energy: https://app.powerbi.com/groups/me/reports/e09268d2-7f5d-4518-bab2-1fd9f39734f2?experience=power-bi ; Space: https://app.powerbi.com/groups/me/reports/388c3a7f-1caf-469b-a779-da2b1d136041?experience=power-bi
Carried from Mid-Term?	☒ No (new / progressed)



EV-12 — Not used

What this proves	No additional evidence claimed.
Deliverable ID (Sec 3)	N/A
Date of Development / Testing	N/A
Verifiable link	N/A
Carried from Mid-Term?	☒ No (not used)

5. Working Demo, Repository & Live Walkthrough Plan

Code repository URL (GitHub/GitLab)	https://github.com/subhikshathirunavukkarasu0503/facilityiq
Final commit ID + date (as of final submission)	ae4a8b4df6493998431db7f1c29786b9c15fd6fa — 20-Sep-2026
Deployed / hosted URL (if any)	https://facilityiq-subhiksha.streamlit.app — currently redirects to Streamlit authentication; local portal remains the demonstrable fallback
Notebook / dashboard / other artefact links	https://github.com/subhikshathirunavukkarasu0503/facilityiq/tree/ae4a8b4df6493998431db7f1c29786b9c15fd6fa/docs/evidence/final

5.1 Live Code Walkthrough Plan

Great Expectations telemetry quality gate	src/facilityiq/qa/data_quality.py; scripts/run_data_quality.py; tests/test_data_quality.py
Browser and load QA	tests/test_selenium_e2e.py; locustfile.py; docs/evidence/final/
Branch	main at ae4a8b4df6493998431db7f1c29786b9c15fd6fa commit
Open in advance	data/lake; models/*_metrics.json; docs/evidence/final/data_quality_report.json; locust_report.html

6. QA Progress (Full Programme)

Test Type (per approved QA strategy)	Tests written / run (total)	Coverage achieved (measured)	Target (per proposal)	Evidence ID(s)
Unit + integration + model + UI regression	80 passed; 3 environment-gated skips	87% measured statement coverage	>80%	EV-08
Great Expectations data quality	14,784 records; 51/51 expectations passed	100% expectation pass; 0 duplicates	Zero invalid records	EV-07

Test Type (per approved QA strategy)	Tests written / run (total)	Coverage achieved (measured)	Target (per proposal)	Evidence ID(s)
Selenium browser E2E	2 journeys: admin five-screen navigation and viewer restriction	2/2 passed	5 E2E scenarios proposed; 2 final browser journeys executed	EV-08
Locust load test	25 users, 30 sec, 1,402 requests	0 failures; p99 5 ms; 47.13 req/s	<2 sec portal response	EV-08

7. Tool & Budget Reconciliation (Full Programme)

Tool / Service (approved)	Approved tier & cost	Used in final solution?	Actual cost (₹)	Reason if changed / not used
Azure IoT Hub	F1 Free / ₹0	No	0	Draft validated to final review, then Azure returned a generic Validation failed; no resource was created.
Azure Functions	Free / ₹0	No	0	Depends on IoT Hub provisioning; not claimed.
Azure Blob Storage	Free/low-cost	Yes	Not independently verified after Mid-Term	Direct lake fallback documented at Mid-Term.
Azure App Service	Free / ₹0	No	0	Plan creation blocked by corporate RBAC; Streamlit fallback.
Databricks + MLflow	Free	Partial	0	Local pandas/scikit-learn; MLflow retained locally.
scikit-learn + PyCaret	Free	Partial	0	scikit-learn used; PyCaret omitted.
Power BI Service	Free / existing organisational licence	Yes	0	Three reports and three semantic models created in My workspace from validated CSV exports.
Streamlit	Free	Yes	0	Five-screen portal and local QA target.
Gemini	Free tier / ₹400 provision	Yes	0 reported	Narratives use cache/fallback.
Ollama + Llama	Free	No	0	Not needed; retained only as planned fallback.
GitHub + Actions	Free	Yes	0	Repository, history and CI workflow.
Pytest + Selenium	Free	Yes	0	80 passing tests; 2 final browser journeys.
Great Expectations	Free	Yes	0	51/51 final data expectations passed.
Domain (optional)	₹450	No	0	Optional purchase omitted.
Groq API	Free tier	Yes, per supplied	0	Included in source;

Tool / Service (approved)	Approved tier & cost	Used in final solution?	Actual cost (₹)	Reason if changed / not used
		project record		claimed L&D condition remains without official feedback.

7.1 Budget Summary

Approved budget ceiling	₹2,500
Actual spend till Mid-Term	₹0 reported in Mid-Term
Actual spend, Mid-Term to Final	₹0 reported; Azure storage billing not independently accessible
Total actual spend	₹0 reported, subject to owner account verification
Buffer remaining	₹2,500 reported, subject to owner account verification

8. Deviations from Approved Proposal (Full Programme)

Item	Approved plan	Actual implementation	Reason for change
IoT ingestion path	IoT Hub → Functions → Blob	Validated local lake → direct Azure Blob fallback; IoT client code retained	Carried from Mid-Term; corporate RBAC blocked provisioning.
ML compute	Databricks + PyCaret	Local pandas/scikit-learn; local MLflow	Carried from Mid-Term; appropriate for POC volume.
Deployment	Azure App Service	Streamlit local/cloud fallback; cloud URL currently auth-gated	Carried from Mid-Term; App Service plan blocked.
Analytics dashboards	Power BI x3	Three live Power BI Service reports backed by validated CSV exports	Completed on 21-Sep-2026; live URLs and screenshots recorded in EV-11.
QA completion	Great Expectations, load and E2E planned	Implemented and executed in final phase	New final-phase completion on 20-Sep-2026.

9. Enhancements & Additional Value-Adds

ID	Enhancement (beyond approved scope)	Why you added it / value it adds	Status	Cost impact (₹)	Evidence ID(s)
EN-01	Four-role access control and admin view	Controls screen access and supports operational demos	Done	0	Mid-Term EV-01
EN-02	Live weather, air quality and footfall context	Adds external operational context	Done	0	Mid-Term EV-02/03
EN-03	Dual AI provider and deterministic cache/template fallback	Keeps explanations available during API/network failure	Done	0	Mid-Term EV-03
EN-04	Automated final data-quality and load evidence	Turns final QA into repeatable scripts and machine-readable reports	Done	0	EV-07, EV-08

10. What Is NOT Completed + Future Scope

Pending item (be specific)	Why it wasn't completed	Flagged at Mid-Term? (Y/N)	Recommended future scope
IoT Hub + Functions live route	Corporate Azure RBAC prevented resource creation	Y	Provision in the assigned RG, set the device connection string, verify route to existing Blob container.
Azure App Service deployment	Corporate RBAC plan prevented creation	Y	Deploy after IT provisions F1 plan; retain Streamlit as backup.
Public cloud portal access	Current Streamlit URL redirects to Streamlit authentication	N	Change sharing settings or grant reviewer access before review.
Official Mid-Term result/feedback and final review dates	Not supplied with project materials	N	Attach official feedback and update Section 1 before submission if available.

11. Risks & Blockers — Final Status

Risk / Blocker	Final Status (Mitigated / Realised / Accepted)	Mitigation taken	Final impact on delivered project
Azure subscription validation blocks IoT Hub/App Service	Accepted	Direct-Blob and local/Streamlit fallbacks retained; the final IoT Hub attempt returned a generic validation failure and created no resource.	Core POC works; approved cloud route is incomplete.
Model accuracy below target	Mitigated	Grouped held-out testing; final metrics retained in repository.	HVAC and motor exceed F1 target; electrical recall is 1.00.
AI/API availability	Mitigated	Caching, provider fallback and deterministic templates.	Portal remains usable without live narrative call.
Power BI report depth	Mitigated	Created three live reports with table visuals and retained validated source datasets.	Reports exist and are evidenced; further visual design remains optional refinement.
Credential exposure in supplied spreadsheet	Realised	Key rotation and password rotation requested; secrets omitted from this document.	Owner must rotate exposed key/passwords before review.

12. Declaration & Pre-Submission Checklist

- ☐ All fields in Section 1 match my L&D Final Decision record exactly, and the Mid-Term document filename is the exact file I uploaded at Week 10.
- ☒ Section 3 lists every deliverable my approved proposal committed to across the full programme (Week 4-17), each with a D-ID and a status, reusing my Mid-Term D-IDs.
- ☒ Deliverables, evidence, and deviations carried forward from Mid-Term are clearly marked as such — nothing is silently omitted.
- ☒ Every "Done" or "Partial" status in Section 3 points to at least one Evidence ID in Section 4 (here or, if carried, in my Mid-Term document).

- ☒ Every evidence block in Section 4 has a specific caption, a Date of Development/Testing, and either a pasted full-size screenshot or a ticked carry-forward reference.
- ☐ The repository link in Section 5 is accessible to the L&D team, the stated final commit exists, and the walkthrough file paths in Section 5.1 exist in the repository.
- ☒ Section 6 coverage figures are measured (tool output attached as evidence), not estimated.
- ☒ Section 7 lists every tool from my approved proposal, including ones I did not use, and reflects final total spend.
- ☒ Section 8 discloses every deviation across the full programme, including ones already reported at Mid-Term.
- ☒ Section 9 (Enhancements) is filled in with cost impact per entry, and its costs reconcile with Section 7 — "None" is written if genuinely not applicable.
- ☒ Section 10 is consistent with Sections 3 and 4 and with my Mid-Term document — no contradictions.
- ☒ I have deleted all grey italic instruction text.
- ☒ I have not renamed, deleted, or reordered any section of this template.
- ☒ Document is saved as [TopicID]_[ParticipantName]_FinalTermDoc.docx and uploaded to Moodle before the deadline.

Declaration: I confirm that all progress claims, evidence, costs, and coverage figures in this document are true and reflect my own individual work across the full programme.

Participant signature / name	Subhiksha Thirunavukkarasu
Date of submission	21-Sep-2026

Submission deadline: before the start of the Week 17 Final Review window, as communicated by L&D. Late or incomplete documentation may affect your final evaluation score and winner eligibility.